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SCALING STI'S SAPPHIRE CRYOCOOLER FOR APPLICATIONS REQUIRING HIGHER HEAT LOADS

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ABSTRACT

Superconductor Technologies Inc. (STI) developed the *Sapphire* cryocooler specifically for the SuperLink[®] product; a high performance superconducting Radio Frequency (RF) front-end receiver used by wireless carriers such as Verizon Wireless and AT&T to improve network cell coverage and data speeds. STI has built and deployed over 6,000 systems operating 24 hours a day (24/7), 7 days a week in the field since 1999. *Sapphire* is an integrated free piston Stirling cycle cryocooler with a cooling capacity of 5 Watts at 77 Kelvin (K) with less than 100 Watts (W) input power. It has a field-proven Mean Time Between Failure (MTBF) of well over 1 million hours, requires zero maintenance and has logged over 250 million cumulative runtime hours.

The *Sapphire* cooler is built on a scalable technology platform, enabling the design of machines with cooling capacities greater than 1 kilowatt (kW). This scalable platform also extends the same outstanding attributes as the *Sapphire* cooler, namely high reliability, zero maintenance, and compact size - all at a competitive cost. This paper will discuss emerging applications requiring higher heat loads and these attributes, describe *Sapphire*, and show a preliminary concept of a scaled machine with a 100 W cooling capacity.

KEYWORDS: Cryogenics, Compressors, Superconducting, Cooling, Scaling

OVERVIEW

There have been significant advancements in the development of High Temperature Superconducting (HTS) wire at STI which will enable new power applications that could significantly reshape the power industry. The emergence of a cost effective, high performance second generation (2G) HTS wire is expected in the next few years, with STI being a key supplier. This wire enables the proliferation of applications such as HTS high

power transmission cables, Superconducting Fault Current Limiters (SCFLs) and HTS electric motors and generators, which could collectively become a significant market. The United States Department of Energy estimates the market size for SFCLs alone to be at several billion dollars over the next 15 years¹. Regardless of the application, however, a cost effective and highly reliable cryogenic ecosystem including the cryocooler is necessary to produce a commercially viable HTS system.

STI has invested over 20 years, not only refining our HTS materials capabilities, but developing cryogenic coolers and thermal management systems. STI's cryogenic coolers are based on the free piston Stirling cycle platform, which STI selected in order to produce highly efficient, supremely reliable and cost effective HTS products for both commercial and government applications. One such product is SuperLink[®]. STI's SuperLink[®] solution is a high performance, superconducting RF front-end receiver used by wireless carriers such as Verizon Wireless to improve network cell coverage and data speeds. For the last 12 years, STI has built and deployed over 6,000 systems, running 24/7.

The cryogenic heart of the system is STI's proprietary *Sapphire* cryocooler. With an MTBF in excess of 1 million hours based on field data, *Sapphire* requires no maintenance and has logged over a quarter of a billion run-time hours. Most importantly, the *Sapphire* cooler is built on a scalable technology platform, enabling the design of machines with cooling capacities greater than 1 kW. This scalable platform also extends the same outstanding attributes of the *Sapphire* cooler; namely high reliability, zero maintenance, compact size, and competitive cost - all crucial to make HTS applications commercially viable.

This paper will discuss new applications for advanced cryogenics requiring higher heat loads that can benefit from STI's scalable *Sapphire* technology. Also, the authors will provide background on *Sapphire's* performance and design attributes and outline a feasibility study which shows a preliminary concept of a scaled machine with a 100 W cooling capacity.

EMERGING HTS APPLICATIONS REQUIRING ADVANCED CRYOGENIC COOLING:

Most applications which require cryogenic cooling today are well known. These applications include: laboratory cryostats, detectors, Magnet Resonance Imaging (MRI), RF communication receivers, medical devices, magnet cooling for high energy physics, semiconductor process equipment, gas liquefaction, etc. Each application varies in cooling capacities, ranging from milliwatts to kilowatts at temperatures from 4 K to 150 K.

A new emerging group of applications could arrive in the next few years. Since the discovery of HTS materials in 1987, the promise of devices such as maglev trains and zero loss power transmission cables has amounted to mostly demonstration projects without an obvious track towards commercial viability. However, recent developments in HTS material technologies have rekindled their prospects. A second generation of HTS Wire is currently being developed at STI that supports appropriate price points and performance levels to enable cost effective HTS power applications. Many of these new superconducting applications are expected to operate in the temperature range between 30 and 77 K, with HTS applications operating at temperatures closer to that of liquid nitrogen (77 K or -196°Celsius). As a general rule of thumb, the complexity and cost of a cryogenic system goes up as temperatures decrease and capacities increase. Therefore, there is an economic driver for newer applications to operate closer to 77 K as efficiently as possible.

As 2G HTS wire improves in performance and cost, a new concern is raised to enable industry acceptance. For large electric superconducting power systems to become a complete reality, a supporting cryogenics ecosystem must also be produced that is cost effective, has low maintenance, very high reliability and tightly integrated with the overall design of the device.

One application that is expected to make a large impact on electric power distribution and transmission are superconducting high power transmission cables. These cables are expected to transmit 3 to 10 times the electrical power of traditional copper or aluminum cables with much improved efficiency. They offer solutions for utilities facing challenges that include: substation footprint availability, lack of available rights of way, and high load connections between substations. They are particularly suited to high power load areas such as the dense urban business districts of large cities, where purchase of an easement and construction costs for traditional low capacity cables would be cost prohibitive. Cooling designs are still in the development stages, but one possibility is a cooling system consisting of large cryocoolers with thermal loads over 1 kW to produce liquid nitrogen, placed at fixed intervals over large distances. These cables would then be cooled by pumping liquid nitrogen through a cryostat built into the cable.

With power demand on the rise and new power generation sources being added, the grid has become overcrowded and vulnerable to catastrophic faults. Faults are abnormal flows of electrical current similar to a short circuit. The unique properties of superconductors enable SCFLs to act as powerful surge protectors, preventing harmful faults from taking down substation equipment, which can result in power loss to businesses and homes. The cooling system of these SFCL's could consist of large cryocoolers generating liquid nitrogen which would then be used to immerse the cable or coils. Thermal loads could range from 100 to 500 Watts.

HTS wire also enables electric motors and generators to operate at higher power densities. When stacked up against a comparable copper wire based electric machine, future superconducting motors and generators could potentially offer a 2 to 3 times size reduction with higher efficiency. A potential sweet-spot for high-powered HTS generators could very well be large, megawatt class offshore wind turbines. Offshore superconducting wind turbines promise to capture clean energy at a lower cost than competing renewables, while delivering power directly to growing coastal cities. The increase in power density provided by superconducting turbines significantly reduces generator weight, maximizes power per tower and increases energy yields, turning wind power into an economically viable alternative.

The cooling system of these motors and generators could consist of large cryocoolers with thermal loads ranging from 100 to 500 Watts to provide liquid nitrogen to immerse the cable or coils or by conduction with a direct connection to a cryocooler. In addition, the location, integration and vibration reduction of the cryocooler may be critical for applications like wind turbines because destructive dynamic interactions between moving components within the pod could occur if not properly done.

BACKGROUND ON STI's LEGACY *SAPPHIRE* CRYOCOOLER

Sapphire is an integral free piston Stirling cycle cryocooler, capable of 5 Watts of heat lift @ 77 K with 100 W input power at room temperature (see FIGURE 1). Its compact design measures 92 millimeters (mm) x 300 mm and weighs only 2.8 kilograms (kg), providing performance levels equivalent to much larger units. It was developed by STI primarily for use in its SuperLink® RF receiver for the telecommunications industry

where long life and MTBF's in the millions are commonplace and expected. In order to compete, STI focused on producing a machine which not only had ultra-high reliability but would be cost effective at the same time. The result is a cooler which requires no maintenance, with an unmatched MTBF of well over 1 million hours based on field data. The cooler was primarily designed for 5 W lift at 77 K but can be operated at different coldfinger temperatures ranging from 40 K and above.

The key features of the *Sapphire* design are shown in FIGURE 2. The unit employs gas bearings, a single piston and displacer, a combination of gas and mechanical springs, efficient heat exchangers, and a passive balancer used to minimize casing vibration.

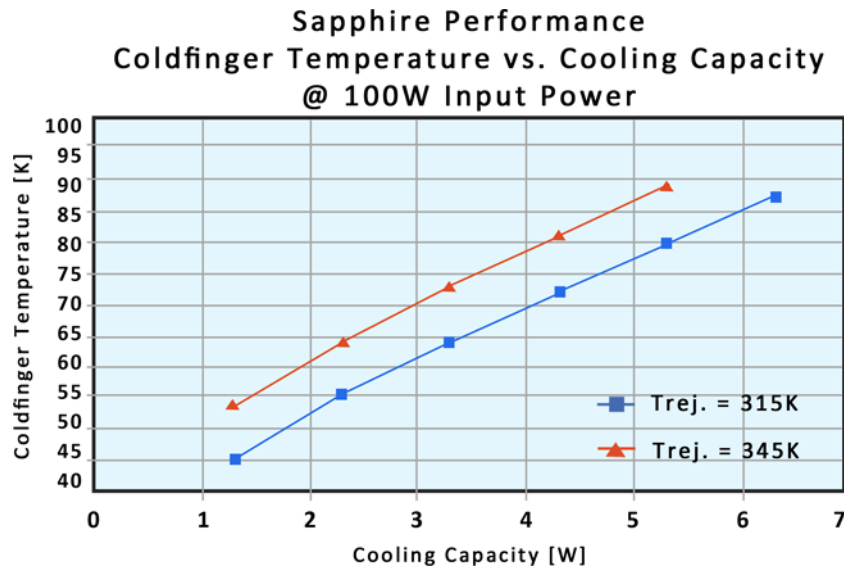


FIGURE 1. *Sapphire* cooler performance.

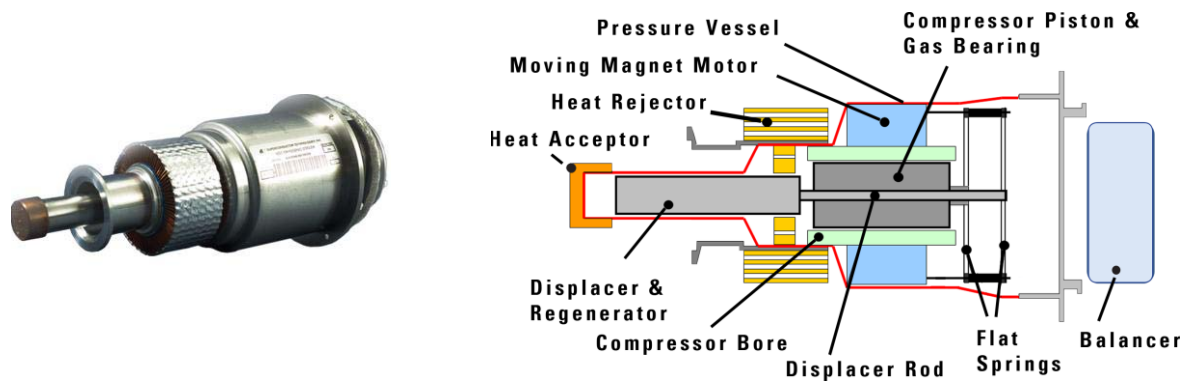


FIGURE 2. *Sapphire* cooler and schematic.

STI builds the *Sapphire* in-house, and over the past 10 years of its production, has continually improved and refined its tools and manufacturing methods to meet its cost objectives without sacrificing quality.

FEASIBILITY STUDY: SCALING THE SAPPHIRE DESIGN FOR HIGHER COOLING CAPACITIES

It has been discussed that there are new potential superconducting applications on the horizon that each require highly reliable, next to zero maintenance, efficient, and low life cycle cost cryogenic systems in order for them to become viable. STI decided to undergo a feasibility study with its main objective being to determine if the *Sapphire* design could be scaled to larger capacities and still retain its key outstanding attributes such as high reliability, zero maintenance, and compact size - all at a competitive cost.

Methodology of Study

The first step was to arrive at a set of initial requirements. A design target of a 100 W cooling capacity at 65 K was chosen because it is within the range of cooling required for motor/generator applications like HTS wind turbines. It is also a large enough increase (30 X) to demonstrate the viability of scaling. Other requirements included an MTBF target of greater than 1 million hours and to be maintenance-free. The cooler had to have optimized vibration levels so it would be easy to integrate in machines like motors and generators without any adverse dynamic effects.

A major goal was to incorporate many of the successful design elements and manufacturing methods from *Sapphire* that helped make it reliable and cost effective into the new design. These include bearing design, effective outgassing/contamination recipes, long life spring design and high efficient moving magnet motor design.

Bearing Design - The bearings chosen are of the same type of gas bearings as used in *Sapphire*. Early analysis of the bearing loads on both the piston and displacer of the 100 W machine indicate that they will be slightly less than the loads experienced in *Sapphire*. For example, the average piston bearing load is about 0.13 N/cm^2 , which is comparable to *Sapphire's* piston bearing load. Based on this, confidence is high that this bearing approach will meet the reliability targets.

Contamination - Outgassing and lack of cleanliness has often been the source for many problems in cryogenic machines. Proper material selection, along with proper fabrication and handling of components is crucial to limiting this problem. For *Sapphire*, STI made careful material selections, and established strong controls and protocols for how parts are fabricated, handled and cleaned at every stage of assembly and test. STI also developed a bake out procedure that not only achieves outstanding results but is cost effective as well. All of these elements were factored in the design of the 100 W cooler.

Springs - A combination of mechanical and gas springs are used in *Sapphire* to achieve the resonance necessary to oscillate the piston and displacer efficiently. For the study, the 100 W concept cooler has similar gas to mechanical spring ratios and uses a similar flat mechanical spring design. STI also used the same proprietary tools that were developed to design and fabricate *Sapphire's* springs. Sufficient design work has been done to indicate that the springs can be, manufacturable, designed for infinite life and be cost effective.

Results of SAGE Calibration with 5W Sapphire

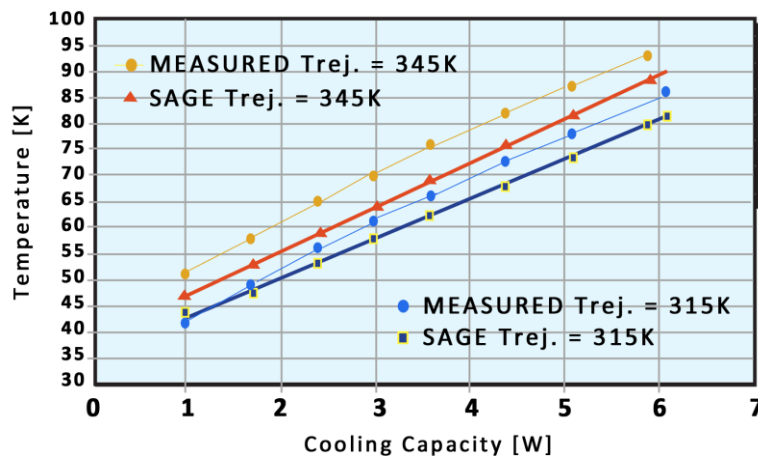


FIGURE 3. SAGE predicted performance vs. *Sapphire* Actual performance and SAGE predicted performance of the 100W unit.

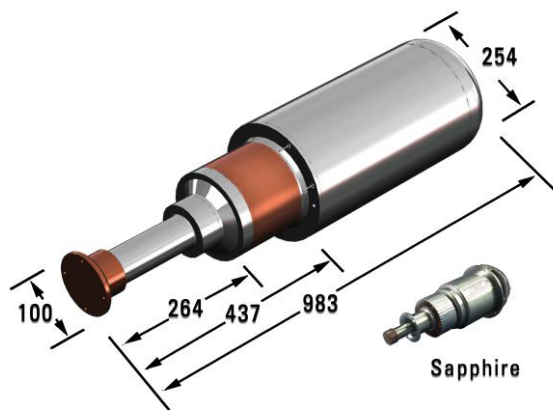
Motor Design - The motor design used for the 100 W concept cooler is the same high efficiency moving magnet type as used in *Sapphire*. STI has over 15 years of experience with this design and has developed some proprietary tools to simulate its performance.

Thermal and mechanical modeling was performed using SAGE simulation software which was developed by Gideon Associates to model and optimize Stirling cycle engines and coolers². In order to gain confidence in simulation software predictions, SAGE was first “calibrated” to the performance of a machine for which all parameters are fully known. *Sapphire* was a perfect candidate for this exercise. FIGURE 3 shows that the results of this “calibration” were within 20% of the actual performance. This is a reasonable result, given the limitations and assumptions built into these types of simulation tools. The final step was to use this “calibrated” simulation, and adjust the parameters and components to meet the performance and design targets of the 100 W concept cooler.

Feasibility Study Results

The resulting preliminary design is shown in FIGURE 4. FIGURE 5 shows the SAGE simulation and its predicted performance. This unit is expected to have a cooling capacity of 100 W at 65 K with an input power of 2.2 kW, which translates into a good overall efficiency of 4.5%. All the key design elements mentioned earlier to make it reliable and robust were successfully incorporated: gas bearings, flat springs, moving magnet motor, and outgassing recipe. Its operating frequency is 60 Hertz and has an estimated weight of about 82 kg.

It has the same basic configuration as the *Sapphire* with one notable exception. For the 100 W concept cooler, an advanced active balancer was placed inside the pressure vessel to optimize vibration instead of a passive balancer mounted outside. This special balancer was developed at STI and is expected to achieve the same low vibration levels as a dual piston/split Stirling configuration. This would enable the cooler to easily integrate into most vibration sensitive application such as large offshore wind turbine generators mentioned earlier.



- Integrated Active Balancer
- Cooling Capacity = 100W @ 65K
- Motor Efficiency > 90%
- Total Efficiency = 4.5%
- Total Input Power = 2.2 kW
- Frequency = 60 Hz
- Charge Pressure = 2.1 Mpa
- Estimated Weight = 82 kg

FIGURE 4. 100 W Concept Cooler size and key results. (dimensions in mm)

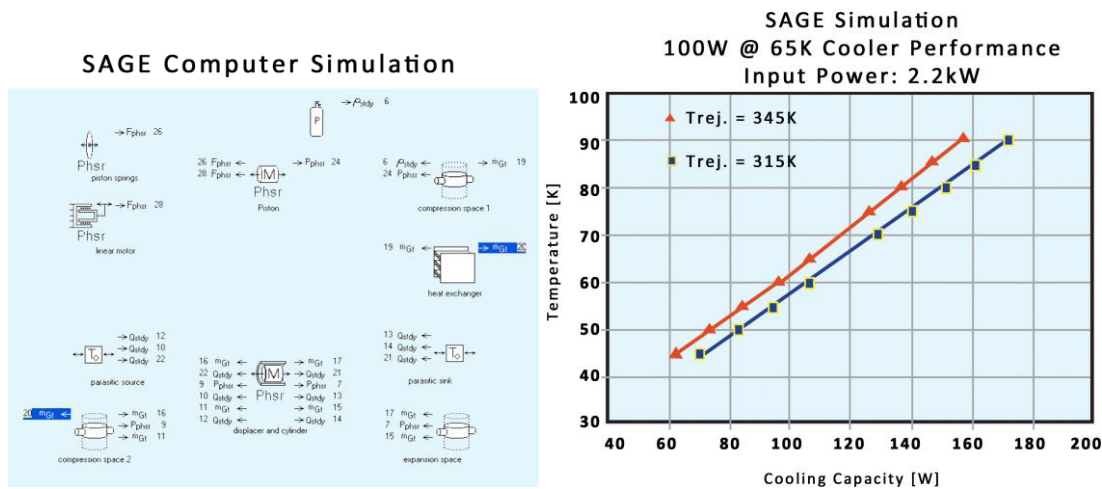


FIGURE 5. SAGE Simulation and predicted performance.

While these initial results indicate that the design is scalable, more work is needed before realizing a working machine. As with any machine of this size and complexity, challenges such as heat exchanger optimization, mechanical alignment and assembly etc. may arise. Given STI's unique experience with these types of machines, these challenges can be met. With the work done in this study, STI is confident that this design approach will not only work for 100 W coolers but also for machines with cooling capacities greater than 1 kW.

SUMMARY

STI is on track to be a key supplier of cost effective, high performance 2G HTS wire in the next few years. This will enable the commercial viability of HTS applications such as superconducting high power transmission cables, superconducting fault current limiters, as well as superconducting motors and generators. Each of these applications requires cryogenic systems and cryocoolers which are very reliable, efficient, maintenance free and cost effective. STI's *Sapphire* cooler has all of these key attributes; however its 5W cooling capacity is not sufficient to meet the needs of these newer applications. A feasibility study, focused on a 100 W @ 65 K cooling capacity, was conducted to determine how the *Sapphire* design approach could be scaled to achieve higher cooling

capacities and retain the key attributes. The results are encouraging and have convinced STI that the platform can be scaled for large cooling capacities while maintaining the ability to be cost effective, efficient and reliable.

ACKNOWLEDGEMENTS

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REFERENCES

1. U.S. Department of Energy, Office of Electric Transmission and Distribution
http://www.oe.energy.gov/DocumentsandMedia/matrix_fault_current_limiter.pdf, Date accessed June 9, 2011.
2. SAGE: Stirling-Cycle Model, Software Package Version 4, Gedeon Associates, Athens, Ohio, 2006.